PP-T-771b
September 20, 1965

SUPERSEDING
Int. Fed. Spec. PP-T-00771a (INT-F&WS)
June 23, 1962 and
Fed. Spec. PP-T-771
March 31, 1931

FEDERAL SPECIFICATION

# TUNA, CANNED

*This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.*

## 1. SCOPE AND CLASSIFICATION

**1.1 Scope.** This specification covers the requirements for the method of preparation, packaging, and packing of canned tuna.

**1.2 Classification.** The product shall be of the following types, forms, colors, and packing media, as specified (see 6.1):

1.2.1 *Types.*

I. Standard (available on the civilian market).

II. Special (made after contract award).

1.2.2 *Forms.*

a. Solid pack.
b. Chunks.
c. Flakes.
d. Grated.

1.2.3 *Packing media.*

(a) In olive oil.
(b) In vegetable oil (other than olive oil).
(c) In water (salt added).
(d) In distilled or demineralized water. This packing media is used for a dietetic pack only. The sodium content of dietetic pack tuna shall not exceed 50 mg. per 100 grams of drained weight. (See 4.2.4.7 and 4.3.1.)

1.2.4 *Color.*

(1) White tuna.
(2) Light tuna.

## 2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

**2.1 Specifications and standards.** The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

*Federal Specification:*

PPP-C-29—Canned Subsistence Items, Packaging and Packing of.

*Federal Standard:*

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the *Index of* Federal Specifications and Standards from established distribution points in their agencies.)

*Military Standards:*

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.



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MIL-STD-668—Minimum Sanitary Standards for Food Establishments.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

**2.2 Other publications.** The following documents form a part of this specification. Unless otherwise indicated, the issues in effect on date of invitation for bids shall apply.

*Governmental*:

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

Federal Food and Drug Administration Definition and Standard of Identity and Standards of Fill of Container for Canned Tuna Fish. Title 21, Chapter 1, Subchapter B, Part 37, Sec. 37.1 and 37.3.

United States Public Health Report 956-1962-Public Health Service Drinking Water Standards.

(Copies may be obtained upon application, accompanied by money order or cash to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402. Prices may be obtained from the Superintendent of Documents.)

*Nongovernmental*:

Official Methods of Analysis of the Association of Official Agricultural Chemists.

(Copies are published by the Association of Official Agricultural Chemists, Post Office Box 540, Benjamin Franklin Station, Washington, D. C., 20044.)

Processes for Low-Acid Canned Foods in Metal Containers, Bulletin No. 26-L Ninth Edition (August 1962), National Canners Association.

(Copies may be obtained from the National Canners Association Research Laboratory, 1133 Twentieth Street, N.W., Washington 6, D. C.)

## 3. REQUIREMENTS

### 3.1 Material.

**3.1.1** *Tuna.* Canned tuna shall be prepared from sound wholesome tuna. The species to be utilized, for the type required, shall comply with provisions of the Federal Food and Drug Administration Definition and Standards of Identity for Canned Tuna.

**3.1.1.1** *Color.*

**3.1.1.1.1** *Color* 1, *white tuna.* White tuna shall be prepared from the Albacore species (*Thunnus germo alalunga*).

**3.1.1.1.2** *Color* 2, *light tuna.* Light tuna shall be prepared from any of the approved species (see 3.1.1).

**3.1.2** *Packing media.*

**3.1.2.1** *Olive oil.* Olive oil shall be the edible oil pressed from the fruit of the olive tree and shall have a typical greenish to light yellow color, shall be free from defects and shall possess a good characteristic odor and flavor.

**3.1.2.2** *Vegetable oil.* Vegetable oil (salad oil) shall be a pure edible vegetable oil (olive oil excepted), and it shall be clear, refined, deodorized oil, and shall possess a good odor and flavor.

**3.1.2.3** *Water.* All water used in canned tuna shall comply with the United States Public Health Service Drinking Water Standards in accordance with U. S. Public Health Report No. 956—1962.

**3.1.3** *Seasonings and flavorings.*

**3.1.3.1** *Salt.* Salt (sodium chloride) shall be of edible grade with or without anticaking agents. The salt shall be clean and free from extraneous material. Iodized salt shall not be used.



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3.1.3.2 *Purified monosodium glutamate and hydrolyzed protein.* Monosodium glutamate and hydrolyzed protein, if used, shall be of edible grade.

3.1.3.3 *Vegetable broth.* Vegetable broth, if used, shall comply with requirements in the Federal Food and Drug Administration Definition and Standard of Identity for Canned Tuna.

3.1.4 *Additives.* Additives, if used, shall comply with the requirements cited in 3.5 as permitted for canned tuna.

**3.2 Types.**

3.2.1 *Type I, standard.* Type I, standard shall be subject to compliance with the requirements specified in 3.3.2, 3.4, and 3.5 and that portion of section 4 applicable to these paragraphs, and applicable parts of section 5 referring to commercial packaging and packing.

3.2.3 *Type II, special.* Type II, special, shall be subject to compliance with all the requirements of the specification.

**3.3 Processing.**

3.3.1 *Preparation of tuna ingredients.* Sound tuna shall be butchered and precooked until the flesh can be easily separated from the bones. The flesh shall be separated from the skin, bones, vascular material, including dark meat and extraneous tissues to the extent that it will comply with end item requirements. Only the white or light tuna portion shall be used. All bones shall be removed to the extent possible. Tuna unfit for canning because of discoloration, staleness, honeycombing, or any other reason shall be excluded.

3.3.2 *Filling, vacuumizing, sealing, and processing of cans.* After the fish have been prepared in accordance with 3.3.1, the meat shall be filled into cans in amounts sufficient to meet the finished product requirements of 3.4 and salt, additives, vegetable broth and water or oil added as applicable. The filled cans shall be vacuumized, sealed, and heat processed (refer to NCA Bulletin No. 26-L for minimum process times) in such a manner as will insure thorough cooking and adequate sterilization of the finished product with a minimum of burning, scorching, or over-cooking. The finished product shall have a vacuum of not less than 1 inch.*

**3.4 Finished product.** The finished product shall have a pleasing appetizing appearance, odor, flavor, and texture whether eaten hot or cold. It shall be reasonably free (see table III for tolerance) from skin, scales, prominent blood streaks, blood clots, bruises including dark meat, and gills, viscera, extraneous tissue, foreign matter, and bones. The product (type, form, color, packing media, seasonings and flavorings, as applicable) shall comply with requirements of the Federal Food and Drug Administration, Definition and Standard of Identity and Standards of Fill of Container for Canned Tuna. The salt (sodium chloride) content shall be not less than 0.5 percent, nor more than 2.2 percent with the exception of the dietetic pack which must contain not more than 50 mg. sodium per 100 grams of drained weight.

3.4.1 *Net weight of product.* The net weight of finished product for given can size shall comply with table IV, in accordance with tolerances given in table I. The lot average net weight shall be not less than contract specified net weight.

3.4.2 *Cans.* The cans finished product shall be coated overall inside with an enamel suitable for the product and shall be sound, clean, and free from rust, serious dents, and seam leaks. Vacuum shall be not less than specified (see 3.3.2) when tested according to the method in PPP-C-29.

* To meet special operational requirements of the Armed Forces the vacuum for military agencies *only* shall be 3 inches.



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**3.5 Federal Food, Drug, and Cosmetic Act.** All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

**3.6 Sanitation.** The processing plant, grounds, equipment, personnel practices, and sanitary practices used in the production of this product shall be such as to minimize the possibility of contamination of the product through microbial growth, dust, pests, rodents, condensate, or other unsanitary sources. The product shall be prepared and processed with minimum delay between the various steps in production.

## 4. SAMPLING, INSPECTION, AND TEST PROCEDURES

**4.1 Sampling.** Samples of the finished product covered by this specification and samples of any materials, components, or constituents entering into the preparation of the finished product may be taken at any time by the Government inspector to determine compliance with the requirements.

**4.2 Inspection.**

4.2.1 *General.* Inspection shall be made by the procuring agency or a duly authorized representative, at the time and place designated by the procuring agency. It may be made at the site of manufacture, both during and after processing and packaging, at any suitable point in transit, and after delivery to the point of destination.

4.2.2 *Factory inspection.* The passing as satisfactory of any detail of processing or material shall not relieve the contractor of responsibility for faulty workmanship or materials which may be discovered at any time prior to final acceptance.

4.2.3 *Final inspection.* Unless otherwise specified, final inspection shall be made before delivery to the point of destination.

4.2.4 *Quality assurance provisions.*

4.2.4.1 *Responsibility for inspection.* The supplier is responsible for meeting all the provisions of the specification prior to submission of supplies to the Government for acceptance. Unless otherwise specified in the contract or order, the supplier is required to perform the examinations and tests prescribed herein, using his own or any other inspection facilities and services acceptable to the Government.

4.2.4.1.1 Inspection records of the examination and tests shall be complete and made available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2.4.2 *Pre-acceptance inspection (plant sanitation inspection).*

4.2.4.2.1 *For civil agencies.* Inspection of plants producing the product will be conducted upon request of the contracting officer to the appropriate government inspection agency. Award of contracts will be limited to establishments known to maintain proper sanitary condition or which have received prior sanitary approval recommended by the inspection agency.

4.2.4.2.2 *For military agencies.* Inspection of plants producing the product shall be conducted by the Military Veterinary Services. Award of contracts shall be limited to establishments which have received prior sanitary approval in accordance with MIL-STD-668.

4.2.4.3 *Acceptance inspection.* Inspections required and determination of acceptability shall be in accordance with provisions set forth in MIL-STD-105, except where otherwise indicated hereinafter.

4.2.4.4 *Inspection of components (examination and testing).* Quality assurance pro-



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visions for components and materials shall be in accordance with 4.2.4.4.1, 4.2.4.4.3, and subsidiary specification to the extent applicable, except that this specification shall govern in the event of conflict.

4.2.4.4.1 *Examination of olive oil, vegetable oil, salt, monosodium glutamate, hydrolyzed protein, and vegetable broth (type II).* Compliance with the requirements of 3.1.2.1, 3.1.2.2, 3.1.3.1, 3.1.3.2, and 3.1.3.3, respectively, shall be satisfied by a review of labels, invoices or similar documents and visual examination. Failure to comply with the stated requirements shall be the basis for rejection of the component.

4.2.4.4.2 *Examination of water (type II).* Examination shall be made to determine that the water used in the production of products under this specification is in compliance with 3.1.2.3. Certified semi-annual laboratory test reports from the state in which the plant is located or of the Federal Government indicating that the water supply conforms to U. S. Public Health Service Drinking Water Standards shall be made available for review. Failure to comply with any of the stated requirements shall be the basis for rejection of the component or the finished product.

4.2.4.4.3 *Examination of tuna fish (type II).* Examination of the tuna shall be performed to determine compliance with the species and color requirements of 3.1.1. Failure to comply with any of the above requirements shall be basis for rejection of the component or the finished product.

4.2.4.5 *Process examination.* Examination shall be made to determine compliance with processing requirements of 3.3.1 and 3.3.2. Records of the formulation and retort time and temperature of each lot shall be maintained by the contractor and made available for review by the inspector. Failure to comply with any of the stated requirements shall be the basis for rejection of the lot.

4.2.4.6 *Examination of end item (both types).* Classification of defects found during examination shall be in accordance with tables I through III and 4.2.4.6.1. The inspection levels, sample units, expression of lot sizes and acceptable quality levels (AQLs) for the above tables shall be as shown below AQLs shall be expressed in defects per hundred units (DHU).

| Table | Inspection level | Sample unit | Lot size expressed in | AQLs | |
|---|---|---|---|---|---|
| | | | | Major | Minor |
| I | S-3 | One primary container | Number of primary containers | — | 4.0 |
| II | S-3 | One primary container | Number of primary containers | 2.5 | |
| III | S-3 | Contents of one primary container | Number of primary containers | 1.0 | 10.0 |

TABLE I—*Examination for net weight*[1]

| Category | Defects |
|---|---|
| *Minor* | |
| 201 | 211x109 and 307x112 or 113 can size. More than 1/4 ounce under contract net weight. |
| 202 | 401x206 can size: More than 1/2 ounce under contract net weight. |
| 203 | 603x408 can size: More than 1 ounce under contract net weight. |

[1] Lot average net weight shall be not less than specified weight.

TABLE II—*Examination for vacuum*

| Category | Defect |
|---|---|
| *Major B* | |
| 151 | Can has less than specified vacuum when tested according to method in Federal Specification PPP-C-29. |



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TABLE III—*Examination of product characteristics*

| Categories | | Defect |
|---|---|---|
| *Major* | *Minor* | |
| | 201 | Presence of numbers of undesirable fish parts of an objectionable size *exceeding* those shown in the following table:[1] |

| *Can Sizes* | *Skin* | *Scales* | *Bones* | *Vascular tissue* |
|---|---|---|---|---|
| 211 x 109 & 307 by 112 or 113 | 1 | 2 | 1 | 1 |
| 401 x 206 | 2 | 4 | 2 | 2 |
| 603 x 408 | 4 | 8 | 4 | 4 |

| | | |
|---|---|---|
| | 202 | Prominent bruise(s), blood clot(s), piece(s) of gills, or viscera, over 1/4 inch in any dimension.[2] |
| 151 | | Type, form, color, or packing media not as specified. |
| 152 | | Foreign material (rust, wood, paper, hair, dirt, insect, or insect parts) either singly or in combination.[3] |
| 153 | | Sour, rancid, stale, scorched, burnt, overcooked, or other objectionable odors or tastes. |
| 154 | | Flesh showing evidence of true honeycombing. |

[1] To classify as objectionable, fish parts must meet the following criteria: Skin—piece is over 1/2-inch in any dimension; scales—over 1/4-inch in two dimensions; bones—any bone or cartilage which is not soft or does not become powdery under firm pressure of a hard object, such as a spoon or spatula, or which is over 1/2-inch in any dimension; vascular tissue includes streaks and dark meat over 1/2-inch in any dimension.

[2] Small blood spots, streaks, and bruises, characteristic of purse-seine caught tuna, are not considered defects.

[3] Presence of paint, chemicals, glass or metal particles, animal excreta, or similar extraneous materials which could harmfully affect or contaminate the product (critical defects) shall be basis for rejection of the entire lot.

**4.2.4.6.1** *Examination for packaging, packing, vacuum, labeling, and marking.* When packaging, packing, vacuum, labeling, and marking are required to be in accordance with PPP-C-29, examination of the packaging, packing, vacuum, labeling, and marking shall be in accordance with the applicable paragraphs and tables of that specification, or of subsidiary specifications when so directed, except that the AQL for can vacuum shall be 10.0 and the vacuum shall be not less than that specified (see 3.3.2).

**4.2.4.7** *Sampling procedure and acceptance criteria for testing of finished product (both types).* Procedures for testing for salt and sodium shall be in accordance with 4.3.1. Requirements shall be applicable to the lot average. The sample unit for finished product testing shall be one primary container. Lot size shall be expressed in primary containers and the sample size for testing shall be the number of sample units indicated by inspection level S-3. Sample units may be composited for testing. Results shall be reported to the nearest unit or decimal specified for the specific requirement and must be within specified limits to be acceptable.

**4.2.4.8** *Standby test samples.* The Government reserves the right to withdraw and hold standby test samples of components or finished product or both (quantity of which shall be not more than twice that of the original Government sample) for inspection purposes. These samples, if not used, will be returned to the vendor.

**4.3 Tests.** Test procedures and controls which differ from those specified herein may be used by the contractor if they provide a quality assurance equivalent to the specified procedures. If the Government inspection activity determines that such procedures and control do not provide, as a minimum, such quality assurance, the contractor will use the test procedure set forth herein. In case of dispute as to test results, the test methods specified herein will govern.

**4.3.1** *Chemical analyses.* Chemical analyses for salt shall be made in accordance with Official Methods of Analysis of the Association of Official Agricultural Chemists; chapter on Fish and other Marine Products—Salt Determination sections.



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Chemical analyses for sodium (used to determine compliance with dietetic pack) shall be made in accordance with the proposed method in volume 47, No. 4, pg. 701, Journal of Official Agricultural Chemists using the wet ashing method.

## 5. PREPARATION FOR DELIVERY

For civil agencies procurement, the definitions and applications of the levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

**5.1 Packaging.** The product shall be packaged in accordance with 5.1.1 or 5.1.2 as specified (see 6.1).

5.1.1 *Level A.* The product, in the net weight and can sizes specified in table IV shall be packaged in metal cans complying with PPP-C-29.

TABLE IV—*Can size and minimum net weight in primary containers*

| Form | Can size | Net weight (min.) (ounces) |
|---|---|---|
| a | 211 by 109 | 3.5 |
| a | 307 by 112 or 113 | 7.0 |
| a | 401 by 206 | 13.0 |
| a | 603 by 408 | 64.0 |
| b | 307 by 112 or 113 | 6.5 |
| b | 401 by 206 | 12.5 |
| b | 603 by 408 | 60.0 |
| c | 307 by 112 or 113 | 6.25 |
| c | 401 by 207 | 12.5 |
| d | 307 by 112 or 113 | 6.25 |
| d | 401 by 206 | 11.75 |

5.1.2 *Level C.* The product shall be packaged in accordance with 5.1.1 and level C requirements of PPP-C-29.

**5.2 Packing, all levels.** The cans of the product shall be packed in accordance with the applicable level as specified (see 6.1) of PPP-C-29.

**5.3 Labeling and marking.** Labeling of cans and marking of shipping containers shall be in accordance with PPP-C-29.

## 6. NOTES

**6.1 Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type, color, form, and packing media required (see 1.2).
(*c*) Applicable levels of packaging and packing (see 5.1 and 5.2).

**6.2 Bid samples.** If for any particular purpose, samples with bids are necessary they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated.

**6.3 Contractor's inspection.** The contractor should be required to assure that the product conforms with the specification prior to submission to the Government for final inspection.

**6.4 Destination inspection.** When the finished product has been inspected and passed at point other than destination the contract should require that the product be inspected at destination for condition and quantity only.

**6.5 Transportation description.** Transportation descriptions and minimum weights applicable to this commodity are:

*Rail*:

Fish, preserved, not otherwise indexed by name.
Carload minimum weight 36,000 pounds.

*Motor*:

Fish, preserved, not otherwise indexed.
Truckload minimum weight 36,000 pounds, subject to Rule 115, National Motor Freight Classification.



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